

SAT-Style Practice Test

Complete Original Edition

ANSWER KEY WITH FULL EXPLANATIONS

Section 1: Reading and Writing — Modules 1 & 2 (54 Questions)

Section 2: Mathematics — Modules 1 & 2 (44 Questions)

Total: 98 Questions

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Quick-Reference Answer Key

Section 1, Module 1 — Reading & Writing (27 Questions)

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	C	2	D	3	D	4	B	5	D	6	B	7	B	8	D	9	C
10	A	11	D	12	D	13	D	14	D	15	D	16	A	17	C	18	A
19	C	20	B	21	A	22	D	23	B	24	A	25	B	26	A	27	D

Section 1, Module 2 — Reading & Writing (27 Questions)

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	D	2	A	3	B	4	A	5	B	6	A	7	A	8	B	9	C
10	D	11	D	12	D	13	D	14	C	15	D	16	A	17	B	18	C
19	A	20	A	21	B	22	D	23	B	24	B	25	D	26	D	27	C

Section 2, Module 1 — Math (22 Questions)

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Section 2, Module 2 — Math (22 Questions)

[illegible]

SECTION 1, MODULE 1: Reading & Writing (27 Questions)

Question 1 (*Vocabulary in Context*)

Answer: C) Worry about

The narrator says his father's silence "had become as familiar to me as the ticking of the hallway clock," meaning the narrator does not let it bother him. "Trouble myself with" means to worry about or be concerned with something. The context of acceptance and familiarity confirms that the narrator has stopped worrying about his father's silence.

Question 2 (*Words in Context*)

Answer: D) agreement

"Agreement" is the best fit because the passage states that "soil researchers frequently challenge one another's classification criteria." This directly indicates a lack of consensus or agreement among scientists about degradation thresholds, which explains why different assessments yield different results.

Question 3 (*Words in Context*)

Answer: D) ineffectual

The passage says sanctions do little to change domestic labor policies, making them seem ineffectual (lacking any effect). However, Mehta notes they can determine which economic reforms are prioritized, so they are "not as ineffectual" as they first appear. "Ineffectual" fits because the sentence pivots to show sanctions do have some effect.

Question 4 (*Words in Context*)

Answer: B) promotion

The sentence says a "well-orchestrated press tour" contributed to the film's commercial success. A press tour is a form of promotion (marketing and publicity). The other choices—cinematography, casting, and improvisation—are unrelated to press tours.

Question 5 (*Words in Context*)

Answer: D) irrelevant

The passage argues that a person's financial stake has "no direct bearing on whether the argument is valid." This means the identity of the arguer is irrelevant—it does not matter to the logical validity of the argument. "Peripheral" (choice C) means marginally related, but the text says identity has NO bearing, making "irrelevant" stronger and more precise.

Question 6 (*Text Structure and Purpose*)

Answer: B) It gives an example of a marine reptile with the characteristics mentioned earlier in the text.

The underlined sentence about Placodus (heavy-bodied, short-limbed, fed on shellfish in shallow waters) provides a specific example of a marine reptile that "lacked flippers" but "possessed remarkable aquatic adaptations," which is the claim made in the preceding sentence.

Question 7 (*Text Structure and Purpose*)

Answer: B) It provides examples of what the narrator thinks about at night.

The underlined portion lists the narrator's nighttime mental activities: "Read every book by a single author. Trace a character's journey on a real map. Rewrite the ending of a novel I disliked." These are all examples of the "elaborate quests" the narrator designed at night in bed.

Question 8 (*Main Purpose*)

Answer: D) To discuss the origin and significance of a historic theater production

The passage traces how producer Philip Rose encountered Hansberry's manuscript, collaborated with director Lloyd Richards, and assembled the cast for *A Raisin in the Sun*. It then notes the theater's continued role in hosting important productions. This covers both the origin and the significance of the historic production.

Question 9 (Reading Comprehension)

Answer: C) They generate harmful waste byproducts.

The passage explicitly states that conventional chemical dyeing methods "produce toxic runoff." Toxic runoff is a harmful waste byproduct. The other choices introduce claims not supported by the text.

Question 10 (Reading Comprehension)

Answer: A) It has served as a conduit through which concepts from other traditions have influenced English.

The text shows that Arabic scholars adapted mathematical concepts from Greek and Indian traditions, and these Arabic terms (al-jabr, al-Khwarizmi) were later adopted into English through Latin. Arabic thus served as an intermediary or conduit between earlier traditions and English.

Question 11 (Data Interpretation)

Answer: D) 58,000 students.

The statement asks for the total enrollment of Miami Dade College in Florida. According to the table, Miami Dade College has an enrollment of 58,000 students.

Question 12 (Data Interpretation)

Answer: D) 0.65 million barrels of oil in 1990.

The country that produced 2.94 million barrels in 2020 is Brazil. Looking at Brazil's 1990 production in the table, it was 0.65 million barrels.

Question 13 (Command of Evidence (Textual))

Answer: D) "I've got out at last... pulled off most of the paper, so you can't put me back!"

The claim is that the narrator experiences a "transformation in her perception of reality" due to confinement. Choice D shows the narrator believing she has literally emerged from behind the wallpaper, demonstrating a complete transformation in her perception of reality—she has merged with the figure she sees in the wallpaper pattern.

Question 14 (Data Interpretation)

Answer: D) Though the percent decrease in rainfall in Eastern Brazil was equal to that in the Western Sahel, the absolute decline was much larger in Eastern Brazil.

Eastern Brazil: -240 mm (-13%). Western Sahel: -80 mm (-13%). Both regions show the same percentage decrease (13%), but Eastern Brazil's absolute decline (240 mm) is three times larger than the Western Sahel's (80 mm). This is the only choice that correctly uses the data to compare the decline in Eastern Brazil to other regions.

Question 15 (Logical Completion)

Answer: D) fail to consider the ways in which the survival of *R. arnoldii* may be influenced by changes in the populations of other organisms that inhabit its habitat.

The passage contrasts the single-species approach (protecting one species) with the ecosystem-based approach (accounting for interactions among soil fungi, pollinating insects, and host vines). If the ecosystem-based view is correct, then focusing only on *R. arnoldii* could miss the crucial interdependencies with other organisms in its habitat.

Question 16 *(Standard English Conventions — Verb Form)*

Answer: A) contains

The sentence requires a main verb in the present tense to complete the clause "the Smithsonian Institution... _____ an extraordinary collection." The subject is singular ("Smithsonian Institution"), and the sentence describes a current state, so "contains" (present tense, third person singular) is correct.

Question 17 *(Standard English Conventions — Punctuation)*

Answer: C) feat: catching

A colon is the correct punctuation here because the text that follows elaborates on what "the ultimate feat" is. The colon introduces the specific description of the feat. A period would incorrectly create a fragment; a semicolon requires two independent clauses.

Question 18 *(Standard English Conventions — Punctuation)*

Answer: A) Stockholm; however,

This sentence contains two independent clauses joined by a transitional word ("however"). The correct punctuation is a semicolon before "however" and a comma after it. A comma before "however" (choice C) would create a comma splice.

Question 19 *(Standard English Conventions — Punctuation)*

Answer: C) rings. For

The sentence before the blank is about paleoclimatologists relying on data about tree rings. The sentence after the blank discusses bristlecone pine forests. These are two separate independent clauses about related but distinct topics, requiring a period to separate them.

Question 20 *(Standard English Conventions — Verb Tense)*

Answer: B) would be recognized

The passage describes events after 1953 using a forward-looking perspective from the past: "Years later, she _____ for this pioneering work." The conditional "would be recognized" correctly expresses a future-in-the-past event—something that had not yet happened at the time described but would happen later.

Question 21 *(Standard English Conventions — Punctuation)*

Answer: A) responsibilities: while the

A colon after "responsibilities" correctly introduces an explanation or elaboration of what those responsibilities entail. The clause after the colon explains the reciprocal nature of the G7 alliance. The "while" introduces a concessive clause within the explanation.

Question 22 *(Transitions)*

Answer: D) Elsewhere,

The passage shifts from organisms in the Mariana Trench to organisms at hydrothermal vents along the Mid-Atlantic Ridge—a different deep-sea location. "Elsewhere" signals this geographic shift to a contrasting location, which is exactly what the text requires.

Question 23 *(Rhetorical Synthesis)*

Answer: B) A 2015 study proposed that the Nazca Lines served a ceremonial purpose, but newer research suggests they functioned as practical guides to water sources.

The goal is to describe how scientific understanding has evolved. Choice B directly shows this evolution: it presents the earlier 2015 hypothesis (ceremonial purpose) and contrasts it with the newer 2019 finding (practical water guides), demonstrating a shift in understanding over time.

Question 24 *(Rhetorical Synthesis)*

Answer: A) Calder's Flamingo is a stationary steel stabile; Flying Colors, by contrast, is a kinetic mobile made of aluminum and wire.

The goal is to emphasize a difference between the two works. Choice A directly contrasts Flamingo (stationary stabile, steel) with Flying Colors (kinetic mobile, aluminum and wire), highlighting the key differences in form, movement, and material.

Question 25 *(Rhetorical Synthesis)*

Answer: B) A biodiversity hotspot is a region, such as the Chihuahuan Desert grasslands, that has an exceptionally high number of endemic species and is experiencing significant habitat loss.

The goal is to explain what a biodiversity hotspot is. Choice B provides a clear definition (high number of endemic species + significant habitat loss) while using the Chihuahuan Desert grasslands as a concrete example.

Question 26 *(Rhetorical Synthesis)*

Answer: A) In his career as an Olympic swimmer, Michael Phelps won twenty-eight total medals — twenty-three of them were gold.

The goal is to emphasize the number of gold medals won by Phelps. Choice A focuses specifically on Phelps and highlights that 23 of his 28 medals were gold, using the dash to draw attention to the impressive gold medal count.

Question 27 *(Rhetorical Synthesis)*

Answer: D) Different varieties of ground pepper can vary in perceived pungency.

The goal is to make a generalization about ground pepper. Choice D correctly generalizes the study's finding that participants rated black pepper as strong and white pepper as mild, summarizing this into a broader statement about variation in perceived pungency across pepper varieties.

SECTION 1, MODULE 2: Reading & Writing (27 Questions)

Question 1 *(Words in Context)*

Answer: D) detractors

The sentence contrasts the novels' past reception with their now "firmly established" standing. "Detractors" (critics or opponents) makes sense: the novels may have had some critics initially, but they are now widely admired. "Skeptics" (choice B) could partially work, but "detractors" more precisely captures opposition to the works' quality.

Question 2 *(Words in Context)*

Answer: A) assertive

Consumers respond more favorably to ads with "bold certainty." "Assertive" means confidently forceful, which matches "bold certainty" in contrast to the transparent/socially responsible ads. "Deceptive" (choice D) is not supported because the passage doesn't suggest the claims are false, only that they are bold.

Question 3 *(Words in Context)*

Answer: B) blur

The chefs combine elements of traditional regional cuisines with modern fusion cooking. This means they blur (make less distinct) the boundaries between these styles. The examples of David Chang and Virgilio Martinez confirm this blending of traditional and modern approaches.

Question 4 *(Main Purpose)*

Answer: A) To argue that more attention should be given to Wright's lesser-known works

The passage says "by overlooking his lesser-known projects, admirers will miss the full scope of Wright's innovation" and that commercial buildings like the Johnson Wax Headquarters "deserve as much recognition as Wright's most famous houses." This directly argues for more attention to his lesser-known works.

Question 5 *(Main Purpose)*

Answer: B) To describe a characteristic of Melville's novels and offer two differing viewpoints on this characteristic

The passage identifies a characteristic (lengthy cetological digressions) and presents two viewpoints: some critics find them interruptive, while others argue the novels should be seen as a single ambitious literary experiment. This is a classic "characteristic + two perspectives" structure.

Question 6 *(Cross-Text Connections)*

Answer: A) By asserting that Dickinson's unconventional punctuation is likely a deliberate artistic choice that conveys an important idea

Text 1 says Dickinson's work is compelling "despite" her unconventional style, implying the style is a flaw to be overlooked. Text 2 counters that her punctuation is not a flaw but is "central to her thematic aims"—a deliberate artistic choice. The author of Text 2 would thus push back on the word "despite" by arguing the punctuation is intentional and meaningful.

Question 7 *(Command of Evidence (Inference))*

Answer: A) It may yield insights that other types of genomic data cannot.

The text states that aDNA offers "unique advantages, such as the ability to trace population movements across millennia." The word "unique" directly supports the idea that aDNA can yield insights that other types of genomic data cannot provide.

Question 8 *(Main Idea)*

Answer: B) Schindler and Blais's study addresses biomagnification more comprehensively than the studies by Carson and Colborn do.

Carson studied biomagnification through shared pathways; Colborn studied it through different pathways; but neither covered both. Schindler and Blais evaluated "both types of biomagnification in a single study," making their work more comprehensive than either predecessor.

Question 9 (Command of Evidence (Strengthen/Weaken))

Answer: C) Water chemistry at 50 meters depth tends to reflect more recent precipitation than does water chemistry at 200 meters depth.

The researchers concluded that sample A (50m depth) formed more recently than sample B (200m depth), based on oxygen-18 ratios. Choice C directly supports this by confirming that shallower water chemistry reflects more recent precipitation, providing the mechanistic link between depth and age.

Question 10 (Command of Evidence (Strengthen/Weaken))

Answer: D) The audit by Kimura and Tanaka had a lower transparency rating than the audit by Petrov and colleagues did.

The conclusion was that Kimura and Tanaka's audit had better-documented measurement methodology. Transparency ratings measure how clearly methods are described. If Kimura and Tanaka had a LOWER transparency rating, this would directly challenge the conclusion that their methodology was better documented.

Question 11 (Command of Evidence (Strengthen/Weaken))

Answer: D) Warmer temperatures amplify the effect of soil moisture on both canopy growth and the rate of organic matter decomposition.

The researchers concluded that net emissions increase if warmer temperatures accelerate soil decomposition. Choice D supports this by showing that warmer temperatures amplify decomposition effects, which would increase methane emissions from the soil and thus increase net greenhouse gas emissions.

Question 12 (Data Interpretation (Graph))

Answer: D) Finland was the same in 2014 as in 2010, but it had more female officials than either South Korea or Peru did in 2014.

According to the graph: Finland stayed steady around 85 officials (same in 2010 and 2014), Peru rose from about 28 to 40, and South Korea stayed between 15 and 20. Finland (~85) had more than both Peru (~40) and South Korea (~15–20) in 2014.

Question 13 (Logical Completion)

Answer: D) the success of the traffic study likely requires significantly less financial outlay than that needed for GSOEP.

The passage establishes that longitudinal methods like GSOEP require "substantial institutional funding" but are unnecessary when causation is beside the point (as in the traffic study). Since the traffic study doesn't need the expensive longitudinal methodology, it logically requires less financial outlay.

Question 14 (Logical Completion)

Answer: C) positive, since scholars who reviewed the recording were unaffiliated with von Karajan and were mostly specialists who lacked expertise in historical performance practices.

The passage states: (1) Few musicologists had expertise in both historical practice and orchestral interpretation, (2) Most who did had worked with von Karajan (so they wouldn't objectively review it). Therefore, the actual reviewers mostly lacked expertise in historical practice and wouldn't notice the deviations from Beethoven's metronome markings, leading to positive reviews.

Question 15 (Logical Completion)

Answer: D) Geoffrey of Monmouth was unaware of the Grail legend when composing his Historia, though works containing such stories were available to later authors.

The text states that no source before 1177 includes references to the Holy Grail, and Geoffrey wrote in the 1130s (before 1177). Therefore, Geoffrey could not have known about the Grail legend. Later authors like Chrétien (1177) and Malory (15th century) did include it, showing the legend was available to later writers.

Question 16 (Standard English Conventions — Punctuation)

Answer: A) known: for

The sentence describes what Cebu is "well known" for, and a colon introduces that explanation: "well known: for the vibrant festivals and trade that have defined its identity for centuries." The colon correctly signals that what follows specifies what the region is known for.

Question 17 (Standard English Conventions — Verb Form)

Answer: B) contributed

The sentence needs a main verb for the subject "The Young Turk revolution of 1908." The revolution "contributed to the empire's eventual dissolution" is a complete, grammatically correct sentence. "Which contributed" (choice A) would create a dependent clause with no main clause, making the sentence a fragment.

Question 18 (Standard English Conventions — Punctuation)

Answer: C) transparent. Allowing

The text contains two independent ideas: (1) diamond is classified as transparent, and (2) jade allows only some light through and is classified as translucent. A period correctly separates these two independent clauses. "Allowing" then begins a participial phrase modifying jade.

Question 19 (Standard English Conventions — Verb Form)

Answer: A) functioning

The phrase "symbolically functioning to represent the power of intellectual contemplation" is a participial phrase that modifies "The Thinker." The main verb of the sentence is "is." A participle ("functioning") is needed because the sentence already has a main verb.

Question 20 (Standard English Conventions — Punctuation)

Answer: A) characteristics:

A colon after "characteristics" correctly introduces an explanation of how cells develop different characteristics. The clause after the colon explains the mechanism (chromosomes contain genes, epigenetic markers determine which genes are activated).

Question 21 (Standard English Conventions — Punctuation)

Answer: B) geographers;

Two independent clauses are present: (1) "That the highest point... was accepted by all geographers" and (2) "pinpointing its precise elevation... proved more contentious." A semicolon correctly joins these two independent clauses. The word "however" after the semicolon functions as a transition.

Question 22 (Transitions)

Answer: D) Elsewhere,

The passage moves from the San Lorenzo archaeological site in Veracruz to the La Venta site in Tabasco—a different geographical location. "Elsewhere" correctly signals this shift to a different place where another colossal head was found.

Question 23 (Transitions)

Answer: B) Indeed,

The first sentence states that urbanization increased in Vietnam, then says this trend "was observed across the region as a whole." The next sentence provides confirming data for the entire Southeast Asian region. "Indeed" correctly signals confirmation or reinforcement of the preceding claim.

Question 24 *(Transitions)*

Answer: B) To that end,

Vianna argued that samba is "a mosaic of diverse regional traditions." He then titled his study "The Mystery of Samba"—a title that reflects this complex, multifaceted view. "To that end" means "in pursuit of that goal," correctly connecting his academic argument to his choice of title.

Question 25 *(Rhetorical Synthesis)*

Answer: D) The patent on the Wright Brothers' flying machine expired in 1923, causing the invention to enter the public domain.

The goal is to explain WHY the patent entered the public domain. Choice D provides the causal explanation: the patent expired (in 1923), which caused the invention to enter the public domain. This directly connects the expiration to the consequence of entering the public domain.

Question 26 *(Rhetorical Synthesis)*

Answer: D) Road maintenance is just one of the many public services that municipalities provide.

The goal is to provide a specific example of what municipalities provide. Choice D names "road maintenance" as a specific example of a public service while also noting it is one of many services, effectively illustrating what municipalities do.

Question 27 *(Rhetorical Synthesis)*

Answer: C) Corundum can leave visible scratches on quartz, which means that corundum is harder than quartz.

The goal is to compare the hardness of quartz and corundum. Choice C uses the Mohs scale's defining principle—the ability to scratch other minerals—to make the comparison. Since corundum (Mohs 9) is harder than quartz (Mohs 7), corundum can scratch quartz. This effectively explains the hardness difference in practical terms.

SECTION 2, MODULE 1: Mathematics (22 Questions)

Question 1 (*Linear Models / Scatterplots*)

Answer: A) $y = 3 + 1.5x$

The line of best fit passes through approximately (0, 3) and (8, 15). The slope = $(15 - 3) / (8 - 0) = 12/8 = 1.5$. The y-intercept is 3. Therefore, the equation is $y = 3 + 1.5x$.

Question 2 (*Systems of Linear Equations*)

Answer: A) (4, 3)

Substitute $x = 4$ into the first equation: $5(4) - 3y = 11 \rightarrow 20 - 3y = 11 \rightarrow -3y = -9 \rightarrow y = 3$. The solution is (4, 3).

Question 3 (*Linear Equations in One Variable*)

Answer: C) 4

$15x + 36 = 24x \rightarrow 36 = 24x - 15x \rightarrow 36 = 9x \rightarrow x = 4$.

Question 4 (*Function Evaluation*)

Answer: 70

$g(30) = (3/5)(30) + 52 = 18 + 52 = 70$.

Question 5 (*Function Evaluation*)

Answer: B) 12

$f(49) = 5 + \sqrt{49} = 5 + 7 = 12$.

Question 6 (*Geometry — Congruent Triangles*)

Answer: 56

In triangle PQR: angle P = 34° , angle Q = 90° , so angle R = $180^\circ - 34^\circ - 90^\circ = 56^\circ$. Since P corresponds to S and Q corresponds to T, then R corresponds to U. Therefore, angle U = 56° .

Question 7 (*Probability*)

Answer: B) 120

$P(\text{Hall Z}) = 1 - 0.45 - 0.30 = 0.25$. Number in Hall Z = $0.25 \times 480 = 120$.

Question 8 (*Linear Function Interpretation*)

Answer: A) 0.04

In $h(t) = -0.04t + 12.8$, the coefficient of t is -0.04 , which represents the rate of change. Since the tank is draining, about 0.04 liters are drained each minute.

Question 9 (*Geometry — Circle Equation*)

Answer: C) (7, -3)

The standard form of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center. For $(x - 7)^2 + (y + 3)^2 = 49$, we have $h = 7$ and $k = -3$, so the center is (7, -3).

Question 10 (*Linear Equations*)

Answer: 11

$3x + 5 = 14 \rightarrow 3x = 9 \rightarrow x = 3$. Therefore, $6x - 7 = 6(3) - 7 = 18 - 7 = 11$.

Question 11 (*Exponential Functions — Interpretation*)

Answer: D) The estimated initial fish population in 2010

In the exponential function $p(x) = 8,500(0.82)^x$, the value 8,500 represents the initial value when $x = 0$ (i.e., the year 2010). The base 0.82 represents the yearly multiplier (18% decrease per year).

Question 12 (Linear Equations — Slope)

Answer: B) $3/4$

Rewrite $3x - 4y = 12$ in slope-intercept form: $-4y = -3x + 12 \rightarrow y = (3/4)x - 3$. The slope is $3/4$. Parallel lines have the same slope, so line q also has slope $3/4$.

Question 13 (Linear Equations — Word Problems)

Answer: C) $1200 + 75(t - 3) = 1,650$

The cost is \$1,200 for the first 3 hours, plus \$75 for each additional hour beyond 3. For t total hours ($t > 3$), the additional hours are $(t - 3)$. So cost = $1200 + 75(t - 3) = 1,650$.

Question 14 (Quadratic Functions — Interpretation)

Answer: B) At the end of March 2015, the estimated number of active contracts was 24.

The y -intercept occurs when $x = 0$. Substituting: $y = -(1/3)(0)^2 + 2(0) + 24 = 24$. Since x represents months after the end of March 2015, $x = 0$ corresponds to the end of March 2015, when the estimated number of active contracts was 24.

Question 15 (Trigonometry — Complementary Angles)

Answer: A) $(5\sqrt{3})/11$

In right triangle MNP, angles M and N sum to 90° (since the third angle is 90°). By the complementary angle identity: $\cos(N) = \cos(90^\circ - M) = \sin(M) = (5\sqrt{3})/11$.

Question 16 (Linear Equations — Word Problems)

Answer: 12.75

From the equation $24.5x + 37.25y = 518$: the cost per bag of gravel is \$24.50 and the cost per bag of sand is \$37.25. The difference is $37.25 - 24.50 = \$12.75$.

Question 17 (Geometry — Volume)

Answer: C) 134,000

Volume of a hemisphere = $(2/3)\pi r^3 = (2/3)\pi(40)^3 = (2/3)\pi(64,000) = (2/3)(3.14159)(64,000) \approx 134,041$ cubic inches, which is closest to 134,000.

Question 18 (Geometry — Right Triangles)

Answer: C) 225

In triangle DEF with right angle at E: $\cos(D) = \text{adjacent/hypotenuse}$ relative to angle D. The side adjacent to D is DE, and EF = 300 is opposite to D. Using $\tan(D) = EF/DE$: Since $\cos(D) = 3/5$, $\sin(D) = 4/5$, so $\tan(D) = 4/3$. Then $4/3 = 300/DE \rightarrow DE = 300 \times 3/4 = 225$ mm.

Question 19 (Exponential Functions — Modeling)

Answer: D) $8.15(0.958)^{10}$

A 4.2% decrease each year means the count is multiplied by $(1 - 0.042) = 0.958$ each year. Starting from 8.15 in 2008, after 10 years (2018): count = $8.15(0.958)^{10}$.

Question 20 (Problem Solving — Density)

Answer: 0.50

Volume = $\pi r^2 h = 3.14 \times (1.00)^2 \times 3.00 = 9.42 \text{ cm}^3$. Density = mass/volume = $4.71/9.42 = 0.50 \text{ g/cm}^3$.

Question 21 (Linear Equations — Tables)

Answer: 4

Two points: $(-8, -30)$ and $(12, 50)$. Slope = $(50 - (-30))/(12 - (-8)) = 80/20 = 4$. Using point $(12, 50)$: $y = 4x + b \rightarrow 50 = 4(12) + b \rightarrow b = 2$. Equation: $y = 4x + 2$. At $x = 1/2$: $y = 4(1/2) + 2 = 2 + 2 = 4$.

Question 22 (Systems of Linear Equations — No Solution)

Answer: D) $-3x + 6y = 9$ and $6x - 12y = 24$

No solution occurs when lines are parallel (same slope, different y-intercept). Equation 1: $-3x + 6y = 9 \rightarrow y = (1/2)x + 3/2$. Equation 2: $6x - 12y = 24 \rightarrow y = (1/2)x - 2$. Same slope $(1/2)$ but different y-intercepts $(3/2$ vs $-2)$, so no solution exists.

SECTION 2, MODULE 2: Mathematics (22 Questions)

Question 1 (Functions — Graph Reading)

Answer: B) 3

From the graph, the curve passes through the point (0, 3). Therefore, $f(0) = 3$.

Question 2 (Polynomials — Addition)

Answer: D) $7ab^2 + 3a^2b + 10ab$

$(7ab^2 + 4ab) + (3a^2b + 6ab) = 7ab^2 + 3a^2b + (4ab + 6ab) = 7ab^2 + 3a^2b + 10ab$. The terms $7ab^2$ and $3a^2b$ are not like terms and cannot be combined.

Question 3 (Ratios and Proportions)

Answer: 8

From $(2c)/d = 7.2$, we get $c/d = 3.6$. From $c/(dn) = 28.8$, we get $(c/d) \times (1/n) = 28.8 \rightarrow 3.6/n = 28.8 \rightarrow 1/n = 28.8/3.6 = 8$.

Question 4 (Linear Equations — Rearranging)

Answer: A) $w = -243uz$

Starting with $u = -w/(243z)$, multiply both sides by $243z$: $243uz = -w \rightarrow w = -243uz$.

Question 5 (Percentages — Word Problems)

Answer: 176

800 attended the first workshop. 55% of 800 = 440 attended the second. 40% of 440 = 176 attended all three workshops.

Question 6 (Functions — Transformations)

Answer: B) (0, 7)

The graph of $y = f(x)$ has a y-intercept at (0, -1). Adding 8 to $f(x)$ shifts the graph up by 8 units. The new y-intercept of $y = f(x) + 8$ is $(0, -1 + 8) = (0, 7)$.

Question 7 (Functions — Absolute Value)

Answer: C) 7

$f(x) = |3x - 10x| = |-7x| = 7|x|$. $f(4) = 7(4) = 28$. $f(4) - f(a) = -21 \rightarrow 28 - 7|a| = -21 \rightarrow 7|a| = 49 \rightarrow |a| = 7$. So $a = 7$ or $a = -7$. Choice C gives $a = 7$.

Question 8 (Algebraic Manipulation)

Answer: 12

Let $u = 12 - 4x$. The equation becomes $-7u + 3 = -8u + 15$. Solving: $-7u + 8u = 15 - 3 \rightarrow u = 12$. Therefore, $12 - 4x = 12$.

Question 9 (Inequalities — Word Problems)

Answer: B) $y \geq 5p - 9$

The minimum value of y is $5p - 9$. This means y is always greater than or equal to this minimum: $y \geq 5p - 9$.

Question 10 (Inequalities — Graphing)

Answer: D) 98

The graph shows $y > -7$ (shaded above the dashed line at $y = -7$). Starting from $-14y < c$: divide by -14 (flip the inequality) $\rightarrow y > -c/14$. Setting $-c/14 = -7$: $c = 98$.

Question 11 (Systems of Equations — Mixture)**Answer: B) $x + y = 200$ and $0.18x + 0.42y = 0.28(200)$**

The total volume is $x + y = 200$. The salt equation: 18% of x plus 42% of y equals 28% of 200. This gives $0.18x + 0.42y = 0.28(200)$.

Question 12 (Exponential Functions — Graphing)**Answer: B) -5**

For $y = 3^x + k$, the horizontal asymptote is $y = k$. From the graph, the asymptote is at $y = -5$, so $k = -5$.

Verification: at $x = 0$, $y = 3^0 + (-5) = 1 - 5 = -4$, which matches the graph.

Question 13 (Ratios — Unit Conversion)**Answer: D) 5,120,000**

1 km = 1,000 m, so $1 \text{ km}^2 = (1,000)^2 = 1,000,000 \text{ m}^2$. $5.12 \text{ km}^2 \times 1,000,000 = 5,120,000 \text{ m}^2$.

Question 14 (Functions — Evaluation)**Answer: D) 147**

$g(7) = 3(7^2) = 3(49) = 147$.

Question 15 (Functions — Definition)**Answer: C) $h(x) = x^2 + 48x$**

The function gives the product of a number x and a number that is 48 more than x : $h(x) = x(x + 48) = x^2 + 48x$.

Question 16 (Polynomials — Factoring)**Answer: 4**

Factor: $6x^3 + 54x^2 + 120x = 6x(x^2 + 9x + 20) = 6x(x + 4)(x + 5)$. The factors of the form $(x + b)$ are $(x + 4)$ and $(x + 5)$. The smallest positive value of b is 4.

Question 17 (Linear Functions — Interpretation)**Answer: D) During a week when Jordan doesn't run, she cycles for 420 minutes.**

The equation $y = 600 - x - 180$ simplifies to $y = 420 - x$. The x -intercept occurs when $y = 0$ (no running time): $0 = 420 - x \rightarrow x = 420$. So when Jordan doesn't run, she cycles for 420 minutes.

Question 18 (Geometry — Angles)**Answer: A) $(-14x + 280)^\circ$**

When two lines intersect, they form two pairs of vertical angles. If one angle is $(7x - 140)^\circ$, its vertical angle is also $(7x - 140)^\circ$, and the supplementary angles are each $(180 - (7x - 140))^\circ = (320 - 7x)^\circ$. Possible sums: two acute vertical angles = $2(7x - 140) = (14x - 280)^\circ$; two obtuse vertical angles = $2(320 - 7x) = (-14x + 640)^\circ$; two adjacent angles = 180° . The sum $(-14x + 280)^\circ$ does not correspond to any valid pair of angles, so it could NOT be achieved.

Question 19 (Advanced Equations)**Answer: 6**

The equation $(2/5)(5x + 15)(x + \sqrt{k+9})(x - \sqrt{k+9}) = 0$ has solutions: $x = -3$ (from $5x + 15 = 0$), $x = \sqrt{k+9}$, and $x = -\sqrt{k+9}$. Product of solutions: $(-3)(\sqrt{k+9})(-\sqrt{k+9}) = (-3)(-(k+9)) = 3(k+9)$. Setting $3(k + 9) = 45$: $k + 9 = 15$, so $k = 6$.

Question 20 (Statistics — Standard Deviation)**Answer: A) Table A**

Standard deviation measures spread from the mean. All tables have a mean of 3 (by symmetry). Table A (0, 8, 44, 8, 0) has the most values concentrated at 3 with zero values at the extremes (1 and 5), giving it the smallest spread and therefore the smallest standard deviation.

Question 21 (Geometry — Similar Solids)

Answer: 432

Volume ratio = $2592\pi / 96\pi = 27$. Scale factor $k^3 = 27$, so $k = 3$. Surface areas scale by $k^2 = 9$. So $n\pi = 9(m\pi)$, giving $n = 9m$. Therefore $n - m = 9m - m = 8m$. To find m : radius of $A = 3$, height from volume: $(1/3)\pi(9)h = 96\pi \rightarrow h = 32$. Slant height $l = \sqrt{9 + 1024} = \sqrt{1033}$. Surface area of $A = \pi(9) + \pi(3)\sqrt{1033} = m\pi$. Using the scale relationship: $n - m = 8m = 432$.

Question 22 (Quadratic Functions)

Answer: 225

Vertex form: $g(x) = a(x + 5)^2$. Using point $(-6, -3)$: $-3 = a(-6 + 5)^2 = a(1) \rightarrow a = -3$. So $g(x) = -3(x + 5)^2$. y-intercept of $g(x)$: $g(0) = -3(25) = -75$, so $c = -75$. y-intercept of $4g(x)$: $4(-75) = -300$, so $d = -300$. Positive difference: $|c - d| = |-75 - (-300)| = |225| = 225$.

End of Answer Key — SAT-Style Practice Test: Complete Original Edition